

System Identification Report

Topic: Dynamic Modelling of a 12-DOF Biped Robot Using Euler-Lagrange Mechanics in PyBullet

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1. System Selection and Approval

The selected system is a lower-body biped robot model implemented in PyBullet and described by a URDF file. It is treated as a nonlinear multi-body dynamic system suitable for dynamic systems analysis, modelling, simulation, and validation through joint-space and contact-force responses.

2. Real-World Application

The system represents a humanoid lower-body platform relevant to legged locomotion, balance control, rehabilitation robotics, humanoid platforms, and dynamic gait research. Such systems must coordinate contact forces, joint torques, link inertia, and center-of-mass motion during tasks such as walking, squatting, and jumping.

3. System Definition

3.1 Inputs

- Joint-space actuation commands applied through PyBullet position control.
- Task-space leg position references used by the inverse-kinematics routine.
- Gait-level inputs such as stride, step height, and support-point progression during walking.
- Terrain inclination through the plane orientation parameter.

3.2 Outputs

- Joint angles, joint torques, and grouped joint stress proxies.
- Ground reaction force magnitude and components.
- Kinetic, potential, and total energy of grouped hip, knee, and ankle subsystems.
- Base pose, link pose, foot trajectories, and center-of-mass trajectory.

3.3 State Variables

For the actuated lower body, a convenient generalized state is defined in joint space as:

$$x = [q_R^T q_L^T \dot{q}_R^T \dot{q}_L^T]^T$$

where q_R and q_L are the six right-leg and six left-leg joint coordinates.

3.4 Key Parameters

- Total robot mass = 5.955684 kg
- Body mass = 4.546 kg
- Right leg mass = 0.704842 kg
- Leg-link lengths used in kinematics: $L1 = 0.18$ m, $L2 = 0.18$ m
- Gravity = 9.81 m/s²
- Per-joint limit range = [-2.5, 2.5] rad, effort limit = 8 N·m, velocity limit = 6 rad/s
- Nominal simulation step = 1/240 s

4. Degrees of Freedom (DOF)

DOF = 12 actuated lower-body degrees of freedom.

- Right leg: hip yaw, hip roll, hip pitch, knee pitch, ankle pitch, ankle roll
- Left leg: hip yaw, hip roll, hip pitch, knee pitch, ankle pitch, ankle roll

This gives full sagittal and lateral lower-body actuation required for standing, walking, squatting, and jumping tasks.

5. Euler-Lagrange Dynamic Model

The project is modelled using Euler-Lagrange mechanics rather than Newton-Euler force balancing. For generalized coordinates q , the Lagrangian is defined as:

$$L = T - V$$

The single-link energies are:

$$T_i = \frac{1}{2} m_i v_{Ci}^T v_{Ci} + \frac{1}{2} \omega_i^T I_i \omega_i$$

$$V_i = m_i g z_{Ci}$$

The equation of motion is expressed in manipulator form as:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + J_c^T \lambda$$

where τ is the actuator torque vector, $J_c^T \lambda$ is the contact-force contribution, and $M(q)$, $C(q, \dot{q})$, $G(q)$ are the mass, Coriolis/centrifugal, and gravitational terms generated by the Euler-Lagrange formulation.

6. Key Assumptions

- All links are rigid bodies with constant mass and inertia.
- Joint friction and motor limits are taken from the URDF and PyBullet setup.
- Ground interaction is concentrated through contact points at the foot links.
- The lower-body model is analysed as an actuated multi-body system in a controlled environment.
- Aerodynamic and structural compliance effects are neglected.
- The analytical model is Euler-Lagrange based; PyBullet supplies the numerical rigid-body simulation.

7. Companion Free Body Diagram

The companion one-leg Euler-Lagrange free-body diagram used for the project is shown below.

FREE BODY DIAGRAM:

